

Yusuf Leblebici

其他

基本信息

姓名: Yusuf Leblebici

所属院校: 洛桑联邦理工学院

地区: Switzerland

年龄范围: 50岁以上

学科领域

微电子科学与工程

集成电路设计与集成系统

研究领域

微电子器件与制造

纳米电子与先进工艺

低功耗电路与能量管理

数字集成电路设计

超大规模集成电路设计

存储器设计与架构

微纳电子与纳米技术

模拟与数模混合电路

模拟与射频IC设计

纳米结构与表征

个人简介

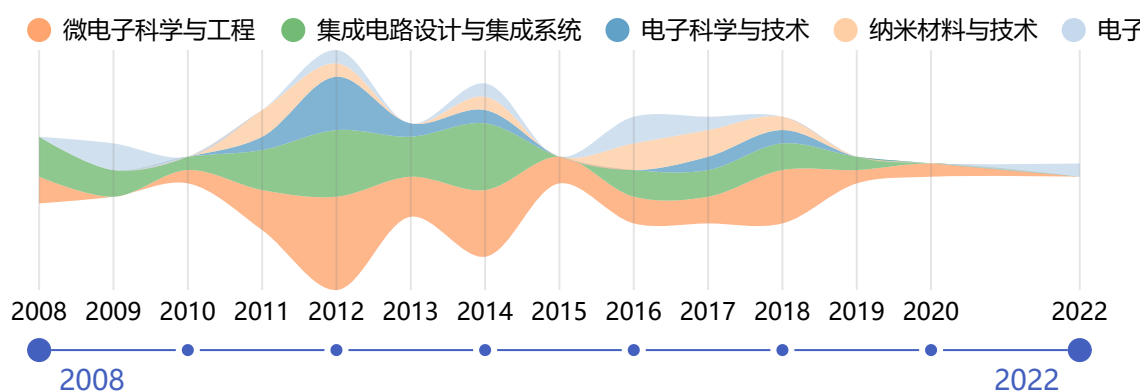
Yusuf Leblebici, Ph.D. from the University of Illinois at Urbana-Champaign in 1990, specializes in modeling and simulation of hot-carrier induced device and circuit degradation for VLSI reliability, classified under Mathematics Subject Classification 94—Information and communication, circuits. His academic mentor was Sung-Mo (Steve) Kang. Dr. Leblebici has supervised numerous doctoral students at École Polytechnique Fédérale de Lausanne (EPFL) and Worcester Polytechnic Institute, contributing significantly to the field of electronics and communication circuits.

教育经历

University of Illinois at Urbana - Champaign - PhD

- 1990

近年发表的论文



A 16-Channel Neural Recording System-on-Chip With CHT Feature Extraction Processor in 65-nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Uran, Arda; Ture, Kerim; Aprile, Cosimo; Trouillet, Alix; Fallegger, Florian; Revol, Emilie C. M.; Emami, Azita; Lacour, Stephanie P.; Dehollain, Catherine; Leblebici, Yusuf; Cevher, Volkan

被引用次数: 7

2022

A 161-mW 56-Gb/s ADC-Based Discrete Multitone Wireline Receiver Data-Path in 14-nm FinFET

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Gain; Kull, Lukas; Luu, Danny; Braendli, Matthias; Menolfi, Christian; Francese, Pier-Andrea; Yueksel, Hazar; Aprile, Cosimo; Morf, Thomas; Kossel, Marcel; Cevrero, Alessandro; Ozkaya, Ilter; Burg, Andreas; Toifl, Thomas; Leblebici, Yusuf

被引用次数: 28

2020

Performance improvement of chip-level CMOS-integrated ReRAM cells through material optimization

MICROELECTRONIC ENGINEERING

Shahrabi, Elmira; LaGrange, Thomas; Demirci, Tugba; Leblebici, Yusuf

被引用次数: 5

2019

Neuromorphic computing with multi-memristive synapses

NATURE COMMUNICATIONS

Boybat, Irem; Le Gallo, Manuel; Nandakumar, S. R.; Moraitis, Timoleon; Parnell, Thomas; Tuma, Tomas; Rajendran, Bipin; Leblebici, Yusuf; Sebastian, Abu; Eleftheriou, Evangelos

被引用次数: 609

2018

Bidirectional Non-Filamentary RRAM as an Analog Neuromorphic Synapse, Part II: Impact of Al/Mo/Pr_{0.7}Ca_{0.3}MnO₃ Device Characteristics on Neural Network Training Accuracy

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Fumarola, Alessandro; Sidler, Severin; Moon, Kibong; Jang, Junwoo; Shelby, Robert M.; Narayanan, Pritish; Leblebici, Yusuf; Hwang, Hyunsang; Burr, Geoffrey W.

被引用次数: 25

2018

FPGA-Based Hardware Implementation of Real-Time Optical Flow Calculation

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY

Seyid, Kerem; Richaud, Andrea; Capoccia, Raffaele; Leblebici, Yusuf

被引用次数: 20

2018

Piezoresistive silicon nanowire resonators as embedded building blocks in thick SOI

JOURNAL OF MICROMECHANICS AND MICROENGINEERING

Esfahani, Mohammad Nasr; Kilinc, Yasin; Karakan, M. Cagatay; Orhan, Ezgi; Hanay, M. Selim; Leblebici, Yusuf; Alaca, B. Erdem

被引用次数: 15

2018

An Eight-Lane 7-Gb/s/pin Source Synchronous Single-Ended RX With Equalization and Far-End Crosstalk Cancellation for Backplane Channels

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Aprile, Cosimo; Cevrero, Alessandro; Francese, Pier Andrea; Menolfi, Christian; Braendli, Matthias; Kossel, Marcel; Morf, Thomas; Kull, Lukas; Oezkaya, Ilter; Leblebici, Yusuf; Cevher, Volkan; Toifl, Thomas

被引用次数: 25

2018

Monolithic Fabrication of Silicon Nanowires Bridging Thick Silicon Structures

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Tasdemir, Zuhail; Peric, Oliver; Sacchetto, Davide; Fantner, Georg Ernest; Leblebici, Yusuf; Alaca, B. Erdem

被引用次数: 8

2018

Neuromorphic computing using non-volatile memory

ADVANCES IN PHYSICS-X

Burr, Geoffrey W.; Shelby, Robert M.; Sebastian, Abu; Kim, Sangbum; Kim, Seyoung; Sidler, Severin; Virwani, Kumar; Ishii, Masatoshi; Narayanan, Pritish; Fumarola, Alessandro; Sanches, Lucas L.; Boybat, Irem; Le Gallo, Manuel; Moon, Kibong; Woo, Jiyoo; Hwang, Hyunsang; Leblebici, Yusuf

被引用次数: 984

2017

Neuromorphic Architecture With 1M Memristive Synapses for Detection of Weakly Correlated Inputs

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Wozniak, Stanislaw; Pantazi, Angeliki; Sidler, Severin; Papandreou, Nikolaos; Leblebici, Yusuf; Eleftheriou, Evangelos

被引用次数: 8

2017

A Monolithic Approach to Downscaling Silicon Piezoresistive Sensors

JOURNAL OF MICROELECTROMECHANICAL SYSTEMS

Esfahani, Mohammad Nasr; Leblebici, Yusuf; Alaca, B. Erdem

被引用次数: 16

2017

Thermal aware design and comparative analysis of a high performance 64-bit adder in FD-SOI and bulk CMOS technologies

INTEGRATION-THE VLSI JOURNAL

Baltaci, Can; Leblebici, Yusuf

被引用次数: 1

2017

Monolithic technology for silicon nanowires in high-topography architectures

MICROELECTRONIC ENGINEERING

Esfahani, M. Nasr; Yilmaz, Mustafa; Wollschlaeger, Nicole; Rangelow, Ivo W.; Leblebici, Yusuf; Alaca, B. Erdem

被引用次数: 11

2017

Top-down technique for scaling to nano in silicon MEMS

JOURNAL OF VACUUM SCIENCE & TECHNOLOGY B

Yilmaz, Mustafa; Kilinc, Yasin; Nadar, Gokhan; Tasdemir, Zuhail; Wollschlaeger, Nicole; Oesterle, Werner; Leblebici, Yusuf; Alaca, B. Erdem

被引用次数: 11

2017

An Insect Eye Inspired Miniaturized Multi-Camera System for Endoscopic Imaging

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Cogal, Omer; Leblebici, Yusuf

被引用次数: 35

2017

Multi-camera platform for panoramic real-time HDR video construction and rendering

JOURNAL OF REAL-TIME IMAGE PROCESSING

Popovic, Vladan; Seyid, Kerem; Pignat, Elieva; Cogal, Omer; Leblebici, Yusuf

被引用次数: 21

2016

Review of advances in neural networks: Neural design technology stack

NEUROCOMPUTING

Almasi, Adela-Diana; Wozniak, Stanislaw; Cristea, Valentin; Leblebici, Yusuf; Engbersen, Ton

被引用次数: 61

2016

Implementation of Low-Power 6-8 b 30-90 GS/s Time-Interleaved ADCs With Optimized Input Bandwidth in 32 nm CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kull, Lukas; Pliva, Jan; Toifl, Thomas; Schmatz, Martin; Francese, Pier Andrea; Menolfi, Christian; Braendli, Matthias; Kossel, Marcel; Morf, Thomas; Andersen, Toke Meyer; Leblebici, Yusuf

被引用次数: 67

2016

A deep etching mechanism for trench-bridging silicon nanowires

NANOTECHNOLOGY

Tasdemir, Zuhail; Wollschlaeger, Nicole; Oesterle, Werner; Leblebici, Yusuf; Alaca, B. Erdem

被引用次数: 20

2016

Determination of the Elastic Behavior of Silicon Nanowires within a Scanning Electron Microscope

JOURNAL OF NANOMATERIALS

Wollschlaeger, Nicole; Tasdemir, Zuhail; Haeusler, Ines; Leblebici, Yusuf; Oesterle, Werner; Alaca, B. Erdem

被引用次数: 4

2016

Design and analysis of jitter-aware low-power and high-speed TSV link for 3D ICs

MICROELECTRONICS JOURNAL

Beanato, Giulia; Gharibdoust, Kiarash; Cevrero, Alessandro; De Micheli, Giovanni; Leblebici, Yusuf

被引用次数: 1

2016

Impact of data serialization over TSVs on routing congestion in 3D-stacked multi-core processors

MICROELECTRONICS JOURNAL

Beanato, Giulia; Cevrero, Alessandro; De Micheli, Giovanni; Leblebici, Yusuf

被引用次数: 4

2016

Hybrid NRZ/Multi-Tone Serial Data Transceiver for Multi-Drop Memory Interfaces

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Gharibdoust, Kiarash; Tajalli, Armin; Leblebici, Yusuf

被引用次数: 25

2015

A Real-Time Multiaperture Omnidirectional Visual Sensor Based on an Interconnected Network of Smart Cameras

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY

Seyid, Kerem; Popovic, Vladan; Cogal, Omer; Akin, Abdulkadir; Afshari, Hossein; Schmid, Alexandre; Leblebici, Yusuf

被引用次数: 10

2015

A comparative experimental investigation on responsivity and response speed of photo-diode and photo-BJT structures integrated in a low-cost standard CMOS process

MICROELECTRONICS JOURNAL

Katic, Nikola; Schmid, Alexandre; Leblebici, Yusuf

被引用次数: 1

2015

Polarity-Controllable Silicon Nanowire Transistors With Dual Threshold Voltages

IEEE TRANSACTIONS ON ELECTRON DEVICES

Zhang, Jian; De Marchi, Michele; Sacchetto, Davide; Gaillardon, Pierre-Emmanuel; Leblebici, Yusuf; De Micheli, Giovanni

被引用次数: 65

2014

Design and characterization of micromachined sensor array integrated with CMOS based optical readout

SENSORS AND ACTUATORS A-PHYSICAL

Erarslan, Refik Burak; Adiyen, Ulas; Lulec, Sevil Zeynep; Olcer, Selim; Temiz, Yuksel; Leblebici, Yusuf; Torun, Hamdi; Urey, Hakan

被引用次数: 9

2014

Memristive Biosensors Under Varying Humidity Conditions

IEEE TRANSACTIONS ON NANOBIOSCIENCE

Puppo, Francesca; Dave, Akshat; Doucey, Marie-Agnes; Sacchetto, Davide; Baj-Rossi, Camilla; Leblebici, Yusuf; De Micheli, Giovanni; Carrara, Sandro

Top-Down Fabrication of Gate-All-Around Vertically Stacked Silicon Nanowire FETs With Controllable Polarity

IEEE TRANSACTIONS ON NANOTECHNOLOGY

De Marchi, Michele; Sacchetto, Davide; Zhang, Jian; Frache, Stefano; Gaillardon, Pierre-Emmanuel; Leblebici, Yusuf; De Micheli, Giovanni

被引用次数: 80

2014

Dynamically adaptive real-time disparity estimation hardware using iterative refinement

INTEGRATION-THE VLSI JOURNAL

Akin, Abdulkadir; Baz, Ipek; Schmid, Alexandre; Leblebici, Yusuf

被引用次数: 13

2014

Configurable Logic Gates Using Polarity-Controlled Silicon Nanowire Gate-All-Around FETs

IEEE ELECTRON DEVICE LETTERS

De Marchi, Michele; Zhang, Jian; Frache, Stefano; Sacchetto, Davide; Gaillardon, Pierre-Emmanuel; Leblebici, Yusuf; De Micheli, Giovanni

被引用次数: 99

2014

MoS2 Transistors Operating at Gigahertz Frequencies

NANO LETTERS

Krasnozhan, Daria; Lembke, Dominik; Nyffeler, Clemens; Leblebici, Yusuf; Kis, Andras

被引用次数: 162

2014

Energy/Reliability Trade-Offs in Low-Voltage ReRAM-Based Non-Volatile Flip-Flop Design

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Kazi, Ibrahim; Meinerzhagen, Pascal; Gaillardon, Pierre-Emmanuel; Sacchetto, Davide; Leblebici, Yusuf; Burg, Andreas; De Micheli, Giovanni

被引用次数: 54

2014

Image Blending in a High Frame Rate FPGA-based Multi-Camera System

JOURNAL OF SIGNAL PROCESSING SYSTEMS FOR SIGNAL IMAGE AND VIDEO TECHNOLOGY

Popovic, Vladan; Seyid, Kerem; Akin, Abdulkadir; Cogal, Oemer; Afshari, Hossein; Schmid, Alexandre; Leblebici, Yusuf

被引用次数: 7

2014

Surface potential compact model for embedded flash devices oriented to IC memory design

SOLID-STATE ELECTRONICS

Garetto, Davide; Rideau, Denis; Gilibert, Fabien; Schmid, Alexandre; Jaouen, Herve; Leblebici, Yusuf

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2013

Applications of Multi-Terminal Memristive Devices: A Review

IEEE CIRCUITS AND SYSTEMS MAGAZINE

Sacchetto, Davide; Gaillardon, Pierre-Emmanuel; Zervas, Michael; Carrara, Sandro; De Micheli, Giovanni; Leblebici, Yusuf

A 3.1 mW 8b 1.2 GS/s Single-Channel Asynchronous SAR ADC With Alternate Comparators for Enhanced Speed in 32 nm Digital SOI CMOS

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kull, Lukas; Toifl, Thomas; Schmatz, Martin; Francese, Pier Andrea; Menolfi, Christian; Braendli, Matthias; Kossel, Marcel; Morf, Thomas; Andersen, Toke Meyer; Leblebici, Yusuf

被引用次数: 204

2013

Post-CMOS Processing and 3-D Integration Based on Dry-Film Lithography

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Temiz, Yuksel; Guiducci, Carlotta; Leblebici, Yusuf

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2013

Ambipolar Gate-Controllable SiNW FETs for Configurable Logic Circuits With Improved Expressive Capability

IEEE ELECTRON DEVICE LETTERS

Sacchetto, Davide; Leblebici, Yusuf; De Micheli, Giovanni

被引用次数: 21

2012

Impact ionization and carrier multiplication in graphene

JOURNAL OF APPLIED PHYSICS

Pirro, Luca; Girdhar, Anuj; Leblebici, Yusuf; Leburton, Jean-Pierre

被引用次数: 28

2012

Analysis of defect capture cross sections using non-radiative multiphonon-assisted trapping model

SOLID-STATE ELECTRONICS

Garetto, Davide; Randriamihaja, Yoann Mamy; Zaka, Alban; Rideau, Denis; Schmid, Alexandre; Jaouen, Herve; Leblebici, Yusuf

被引用次数: 12

2012

Multiterminal Memristive Nanowire Devices for Logic and Memory Applications: A Review

PROCEEDINGS OF THE IEEE

Sacchetto, Davide; De Micheli, Giovanni; Leblebici, Yusuf

被引用次数: 40

2012

Power and area efficient MOSFET-C filter for very low frequency applications

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Tajalli, Armin; Leblebici, Yusuf

被引用次数: 7

2012

Vertically-stacked gate-all-around polysilicon nanowire FETs with sub- μm gates patterned by nanostencil lithography

MICROELECTRONIC ENGINEERING

Sacchetto, Davide; Xie, Shenqi; Savu, Veronica; Zervas, Michael; De Micheli, Giovanni; Brugger, Juergen; Leblebici, Yusuf

被引用次数: 9

2012

Resistive Programmable Through-Silicon Vias for Reconfigurable 3-D Fabrics

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Sacchetto, Davide; Zervas, Michael; Temiz, Yuksel; De Micheli, Giovanni; Leblebici, Yusuf

被引用次数: 22

2012

A Spherical Multi-camera System with Real-time Omnidirectional Video Acquisition Capability

IEEE TRANSACTIONS ON CONSUMER ELECTRONICS

Afshari, Hossein; Popovic, Vladan; Tasci, Tugce; Schmid, Alexandre; Leblebici, Yusuf

被引用次数: 21

2012

Design and Testing Strategies for Modular 3-D-Multiprocessor Systems Using Die-Level Through Silicon Via Technology

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Beanato, Giulia; Giovannini, Paolo; Cevrero, Alessandro; Athanasopoulos, Panagiotis; Zervas, Michael; Temiz, Yuksel; Leblebici, Yusuf

被引用次数: 19

2012

Modeling Stressed MOS Oxides Using a Multiphonon-Assisted Quantum Approach-Part I: Impedance Analysis

IEEE TRANSACTIONS ON ELECTRON DEVICES

Garetto, Davide; Randriamihaja, Yoann Mamy; Rideau, Denis; Zaka, Alban; Schmid, Alexandre; Leblebici, Yusuf; Jaouen, Herve

被引用次数: 5

2012

Modeling Stressed MOS Oxides Using a Multiphonon-Assisted Quantum Approach-Part II: Transient Effects

IEEE TRANSACTIONS ON ELECTRON DEVICES

Garetto, Davide; Randriamihaja, Yoann Mamy; Rideau, Denis; Zaka, Alban; Schmid, Alexandre; Leblebici, Yusuf; Jaouen, Herve

被引用次数: 12

2012

A comparative study on fabrication techniques for on-chip microelectrodes

LAB ON A CHIP

Temiz, Yuksel; Ferretti, Anna; Leblebici, Yusuf; Guiducci, Carlotta

被引用次数: 30

2012

Ambipolar silicon nanowire FETs with stenciled-deposited metal gate

MICROELECTRONIC ENGINEERING

Sacchetto, Davide; Savu, Veronica; De Micheli, Giovanni; Brugger, Juergen; Leblebici, Yusuf

被引用次数: 7

2011

Amplitude modulation based readout for very dense active microelectrode arrays

IEICE ELECTRONICS EXPRESS

Joye, Neil; Schmid, Alexandre; Leblebici, Yusuf

被引用次数: 0

2011

Monolithic integration of Si nanowires with metallic electrodes: NEMS resonator and switch applications

JOURNAL OF MICROMECHANICS AND MICROENGINEERING

Arkan, Evren F.; Sacchetto, Davide; Yildiz, Izzet; Leblebici, Yusuf; Alaca, B. Erdem

被引用次数: 22

2011

Silicon Nanowire Arrays and Crossbars: Top-Down Fabrication Techniques and Circuit Applications

SCIENCE OF ADVANCED MATERIALS

Ben-Jamaa, M. Haykel; Gaillardon, Pierre-Emmanuel; Clermidy, Fabien; O'Connor, Ian; Sacchetto, Davide; De Micheli, Giovanni; Leblebici, Yusuf

被引用次数: 9

2011

Low-Power and Widely Tunable Linearized Biquadratic Low-Pass Transconductor-C Filter

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Tajalli, Armin; Leblebici, Yusuf

被引用次数: 13

2011

Top-down fabrication of very-high density vertically stacked silicon nanowire arrays with low temperature budget

MICROELECTRONIC ENGINEERING

Zervas, Michael; Sacchetto, Davide; De Micheli, Giovanni; Leblebici, Yusuf

被引用次数: 29

2011

Polysilicon Nanowire Transistors and Arrays Fabricated With the Multispacer Technique

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Ben Jamaa, M. Haykel; Cerofolini, Gianfranco; De Micheli, Giovanni; Leblebici, Yusuf

被引用次数: 5

2011

Optimization of the wire grid size for differential routing: Analysis and impact on the power-delay-area tradeoff

MICROELECTRONICS JOURNAL

Alioto, Massimo; Badel, Stephane; Leblebici, Yusuf

被引用次数: 2

2010

Slew Controlled LVDS Output Driver Circuit in 0.18 μm CMOS Technology

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Tajalli, Armin; Leblebici, Yusuf

被引用次数: 47

2009

Ultra-low power 32-bit pipelined adder using subthreshold source-coupled logic with 5 fJ/stage PDP

MICROELECTRONICS JOURNAL

Tajalli, Armin; Brauer, Elizabeth J.; Leblebici, Yusuf

被引用次数: 9

2009

Electrical modeling of the cell-electrode interface for recording neural activity from high-density microelectrode arrays

NEUROCOMPUTING

Joye, Neil; Schmid, Alexandre; Leblebici, Yusuf

被引用次数: 62

2009

On heat transfer at microscale with implications for microactuator design

JOURNAL OF MICROMECHANICS AND MICROENGINEERING

Ozsun, Ozgur; Alaca, B. Erdem; Yalcinkaya, Arda D.; Yilmaz, Mehmet; Zervas, Michalis; Leblebici, Yusuf

被引用次数: 33

2009

Field Programmable Compressor Trees: Acceleration of Multi-Input Addition on FPGAs

ACM TRANSACTIONS ON RECONFIGURABLE TECHNOLOGY AND SYSTEMS

Cevrero, Alessandro; Athanasopoulos, Panagiotis; Parandeh-Afshar, Hadi; Verma, Ajay K.; Attarzadeh-Niaki, Seyed-Hosein; Nicopoulos, Chrysostomos; Gurkaynak, Frank K.; Brisk, Philip; Leblebici, Yusuf; lenne, Paolo

被引用次数: 7

2009

Leakage Current Reduction Using Subthreshold Source-Coupled Logic

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Tajalli, Armin; Leblebici, Yusuf

被引用次数: 12

2009

CMOS realization of two-dimensional mixed analog-digital Hamming distance discriminator circuits for real-time imaging applications

MICROELECTRONICS JOURNAL

Badel, Stephane; Schmid, Alexandre; Leblebici, Yusuf

被引用次数: 4

2008

Variability-Aware Design of Multilevel Logic Decoders for Nanoscale Crossbar Memories

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Jamaa, Mohamed Haykel Ben; Moselund, Kirsten Emile; Atienza, David; Bouvet, Didier; Ionescu, Adrian Mihai; Leblebici, Yusuf; De Micheli, Giovanni

被引用次数: 5

2008

Subthreshold source-coupled logic circuits for ultra-low-power applications

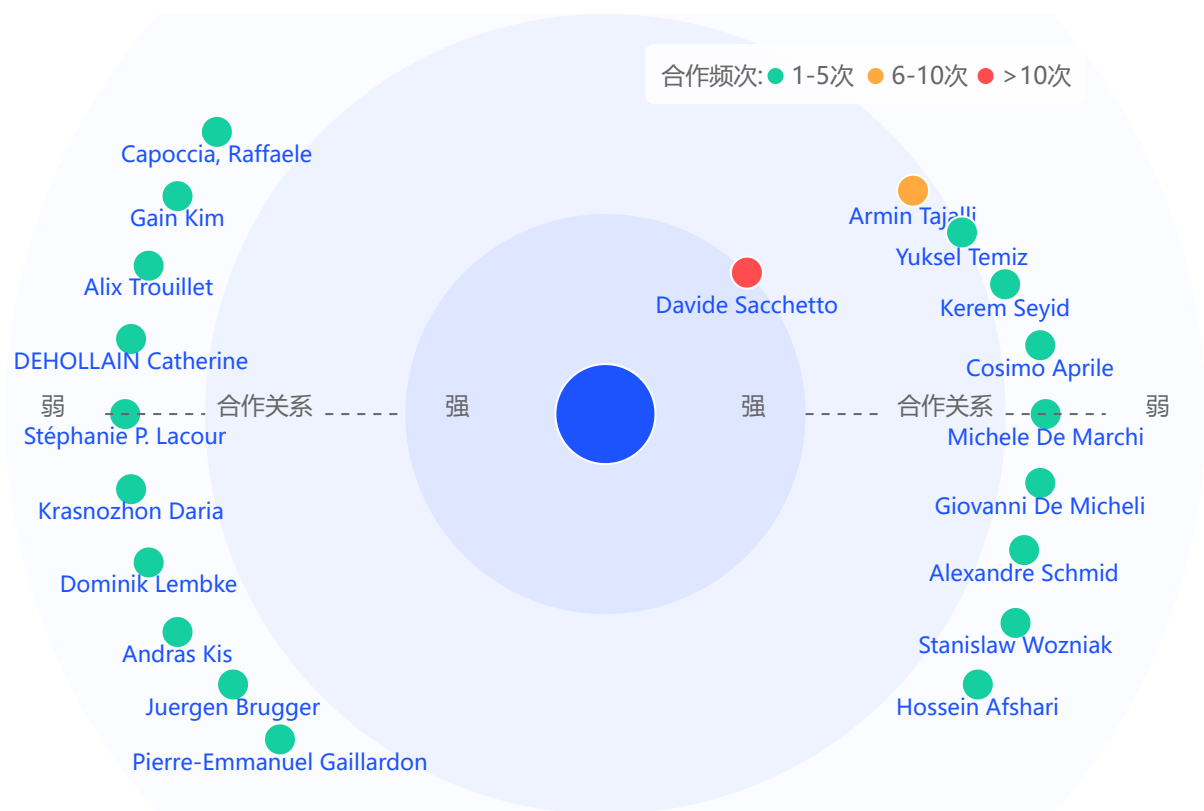
IEEE JOURNAL OF SOLID-STATE CIRCUITS

Tajalli, Armin; Brauer, Elizabeth J.; Leblebici, Yusuf; Vittoz, Eric

被引用次数: 78

2008

学术合作



Davide Sacchetto

洛桑联邦理工学院

合作次数: 15

Armin Tajalli

洛桑联邦理工学院

合作次数: 6

Yuksel Temiz

洛桑联邦理工学院

合作次数: 5

Kerem Seyid

洛桑联邦理工学院

合作次数: 4

Cosimo Aprile

洛桑联邦理工学院

合作次数: 3

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